Economia das Organizações e dos Incentivos — MPE Insper

Exercise List: Aulas 1–2: Complementarities, Value Maximization and Bargaining (graduate level)

Instructions. Multiple-choice items: mark the single correct option (no penalty for errors). *True/False* items: every 2 wrong items cancel 1 correct one (score = $\max\{0, \text{correct} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step; a result the item asks you to derive may not be quoted. All questions are symbolic: state the admissible range of every parameter you use. Throughout, "quasi-linear" means $u_i = m_i + (\text{money transfers}) + v_i(\cdot)$, i.e. no wealth effects in the sense of Milgrom & Roberts (1992, p. 35).

Block 1 — Complementarities and coherent systems *(Aula 1, with Aula 2 • 25 points)*

Question 1. (10 points) Objective items.

a) (3 points) *Complementarities × value maximization.* At a Brazilian retail bank, one director sets the degree of delegation to branch managers, x, and another the intensity of incentive pay, y. Bank profit is $\pi(x, y) = tx + sy + bxy - \frac{c_1}{2}x^2 - \frac{c_2}{2}y^2$, with $b > 0$ and $c_1 c_2 > b^2$. The two directors have quasi-linear preferences, can sign enforceable agreements on (x, y), make unrestricted side payments and bargain efficiently. Better local information raises t. The total effect $\mathrm{d}y^*/\mathrm{d}t$ on the agreed incentive intensity is:

A. $b/(c_1 c_2)$

B. $c_2/(c_1 c_2 - b^2)$

C. $b/(c_1 c_2 - b^2)$

D. $-b/(c_1 c_2 + b^2)$

E. $b/(c_2^2 - b^2)$

b) (3 points) A textile firm in Santa Catarina weighs three lean practices $J, T, S \in \{0, 1\}$. Relative to the status quo $(0, 0, 0)$, profit is $\pi = -\alpha(J + T + S) + \beta(JT + JS + TS)$ with $\alpha, \beta > 0$. The set of parameters for which (i) adopting all three is the unique global maximum, (ii) every single first step loses money and (iii) every two-practice corner is below the status quo, is:

A. $\alpha < \beta < 2\alpha$

B. $\beta > \alpha$

C. $\alpha/2 < \beta < \alpha$

D. $\beta > 2\alpha$

E. $\alpha < \beta < 3\alpha$

c) (2 points, T or F) *Complementarities × Coase.* Two managers bargain efficiently, with quasi-linear preferences and unrestricted transfers, over two *substitute* practices: $\pi(x, y; t)$ is strictly concave, $\pi_{xy} < 0$, $\pi_{xt} > 0$ and $\pi_{yt} = 0$. Then the agreed x responds to t by *less* than it would if y were held fixed.

d) (2 points, T or F) Let $f(x, t)$ have increasing differences in (x, t) on $[0, 1] \times \mathbb{R}$ and be continuous in x. Then the largest maximizer $\bar{x}(t)$ of $f(\cdot, t)$ on $[0, 1]$ is non-decreasing in t, even if f is not concave in x.

Question 2. (15 points) *Complementarities (Aula 1) × value maximization and Coase (Aula 2).* An auto-parts maker in the ABC paulista is converting a plant to a lean system. There are $n \geq 2$ practices (just-in-time delivery, multi-skilled teams, in-line quality control, ...), each run by a different department head; practice i is used at level $x_i \geq 0$. Plant profit, net of all the heads' private costs, is

$$\pi(x) = \sum_{i=1}^{n} t_i x_i + \frac{b}{2} \sum_{i \neq j} x_i x_j - \frac{c}{2} \sum_{i=1}^{n} x_i^2, \qquad t_i > 0,\ c > 0,\ b \geq 0.$$

The heads have quasi-linear preferences, can sign enforceable agreements on every x_i, make unrestricted transfers among themselves, and bargain efficiently. Unless stated otherwise, $t_i = t$ for all i.

a) (3 points) Explain why the heads choose x to maximize π whatever their bargaining weights. Find the necessary and sufficient condition on (n, b, c) for π to have a unique maximizer, and describe what happens when it fails.
Resolution:

b) (3 points) Derive the optimal levels x^* and the maximal profit π^*. Prove that the optimum is symmetric.
Resolution:

c) (3 points) The return to practice 1 rises (t_1 increases, the other t_i fixed). Derive $\partial x_1^*/\partial t_1$ and $\partial x_j^*/\partial t_1$ for $j \neq 1$, and compare the own response with the response obtained when all other practices are held fixed. For $n = 2$, draw the two best-reply lines in the (x_1, x_2) plane, the optimum before and after the shock, and the rounds of adjustment in which the two heads take turns re-optimizing.
Resolution:

d) (3 points) A consultant proposes evaluating the system as n separate pilot projects: in each pilot one practice is set optimally while all others stay at zero. Define the *system premium* P as π^* minus the sum of the n pilots' profits. Derive P in closed form, prove that $P > 0$ iff $b > 0$, and find the share of π^* that the pilots miss.
Resolution:

e) (3 points) Parameter analysis. Prove the sign of the effect of t, b, c and n (treated as continuous) on x_i^*, on profit per practice π^*/n, on the multiplier m_n and on the share P/π^*. Summarize in a table, study the limits $b \to 0$ and $(n-1)b \to c^-$, and give the economic reading.
Resolution:

Block 2 — Value maximization, the Coase theorem and the division of surplus

(Aula 2 • 25 points)

Question 3. (10 points) Objective items.
a) (3 points) A Petrobras technician chooses between two shift assignments y_1, y_2 (M&R's overtime example, p. 38); $x > 0$ is money and $v(\cdot)$ is a real-valued function. Ignoring the ability-to-pay condition (wealth is large), which utility function displays *no wealth effects* in M&R's sense (p. 35)?

A. $u(x, y) = \ln x + v(y)$

B. $u(x, y) = x\, e^{v(y)}$

C. $u(x, y) = x + v(y)\sqrt{x}$

D. $u(x, y) = \big(x + v(y)\big)^3$

E. $u(x, y) = v(y) \ln x$, with $v > 0$

b) (3 points) In Cubatão (SP) a chemical plant gains $G > 0$ from discharging effluent into a river; the neighbourhood downstream, with wealth M, has utility $u = 2\sqrt{x} + \theta c$, where $c \in \{0, 1\}$ indicates a clean river and $0 < \theta < 2\sqrt{M}$. Bargaining is costless and information complete. The set of values of G for which the assignment of the right (to the plant or to the neighbourhood) decides whether discharging occurs is:

A. empty (the Coase theorem applies whatever the utility function)

B. $(\theta\sqrt{M} - \theta^2/4,\ \theta\sqrt{M} + \theta^2/4)$

C. $(\theta\sqrt{M} - \theta^2/2,\ \theta\sqrt{M} + \theta^2/2)$

D. $(\theta\sqrt{M},\ \theta\sqrt{M} + \theta^2/4)$

E. $(2\theta\sqrt{M} - \theta^2/2,\ 2\theta\sqrt{M} + \theta^2/2)$

c) (2 points, T or F) *Coase × complementarities.* Two divisions with quasi-linear preferences control complementary practices: division 1 sets x, division 2 sets y, joint profit $\pi = tx + sy + bxy - \frac{c_1}{2}x^2 - \frac{c_2}{2}y^2$ with $t, s, b > 0$, $c_1 c_2 > b^2$. They contract on (x, y) with unrestricted transfers and bargain efficiently. A rise in division 1's cost parameter c_1 lowers the agreed level of y.

d) (2 points, T or F) *Bargaining × complementarities.* In the setting of item c), the agreement is the generalized Nash bargaining solution with weight $\beta \in (0, 1)$ for division 1 and threat points (d_1, d_2). A better outside option for division 1 (higher d_1) raises both agreed levels x and y, because complements move together.

Question 4. (15 points) *Bargaining (Nash solution, threat points) × efficiency (value maximization, wealth effects).* On a river of the São Francisco basin, an irrigation company U upstream chooses its withdrawal $y \geq 0$; a fishing cooperative D downstream is harmed. U's gross benefit is $B(y) = \alpha y - \frac{1}{2}y^2$; D's harm in money is $\frac{\kappa}{2}y^2$, with $\alpha > 0$, $\kappa > 0$. The water agency (ANA) gives the right either to D (*regime D*: no withdrawal without D's consent) or to U (*regime U*: unrestricted withdrawal). Both parties observe y and all parameters, can contract on y and on a transfer, and the agreement is the generalized Nash bargaining solution with weight $\beta \in (0, 1)$ for U; each regime's threat point is the outcome without agreement. Until item d), preferences are $u_U = m_U + B(y) + x_U$, $u_D = m_D - \frac{\kappa}{2}y^2 + x_D$, with $x_U + x_D = 0$.

a) (3 points) Prove that the Pareto-efficient allocations are exactly those with one particular withdrawal level, and find it.
Resolution:

b) (3 points) Characterize the bargaining outcome in each regime: threat points, withdrawal, the transfer (with its direction) and both payoffs. What does the comparison show?
Resolution:

c) (3 points) In the plane of payoffs net of m_i, (u_U, u_D), draw the utility possibility frontier, the threat points of both regimes and both bargaining outcomes. Read the Coase theorem off the figure.
Resolution:

d) (3 points) Now D's utility is $h(m_D + x_D) - \frac{\kappa}{2}y^2$, with $h' > 0$, $h'' < 0$ (U unchanged). Characterize the Pareto frontier: show that along it the efficient withdrawal is a strictly decreasing function of D's utility, and give the case $h(w) = \ln w$. Use this to show that, if every agreement U accepts in

regime U leaves D worse off than $h(m_D)$, then every agreement of regime U has a strictly larger withdrawal than every agreement of regime D. Which special case of h restores item a)?
Resolution:

e) (3 points) Parameter analysis for items a)–b). Prove the sign of the effect of α, κ and β on y^*, W^*, S_U, τ_D and τ_U; where a sign is ambiguous, find the threshold. Summarize in a table, study $\kappa \to 0$, $\kappa \to \infty$, $\beta \to 0$ and $\beta \to 1$, and give the economic reading, including what happens to regime U when D's wealth is limited.
Resolution:

Block 3 — Integrative: complementary investments, Coase and alternating offers

(Aulas 1–2 + MWG 9.AA • 25 points)

Common setting for Questions 5 and 6. In an industrial condominium on the model of GM's Gravataí (RS) plant, an assembler A chooses $x \geq 0$ (just-in-time integration) and a co-located system supplier S chooses $y \geq 0$ (dedicated capacity). Joint revenue is $R(x, y) = t(x + y) + bxy$; private costs are $\frac{c}{2}x^2$ for A and $\frac{c}{2}y^2$ for S, with $t > 0$ and $c > b > 0$. Preferences are quasi-linear.

- *Regime I.* x and y are contractible and transfers unrestricted. The parties bargain by alternating offers, A proposing first; an offer is (x, y, τ), where A receives R and pays τ to S. Discount factors $\delta_A, \delta_S \in (0, 1)$; without agreement each gets 0 per period.
- *Regime II.* No transfers are possible: an outside rule gives A the share σ of R and S the share $1 - \sigma$. x and y are not contractible and are chosen simultaneously.

Question 5. (10 points) Objective items.
a) (3 points) In regime I, suppose bargaining lasts at most three rounds (A proposes in rounds 1 and 3, S in round 2; after round 3 both get 0). In the subgame-perfect equilibrium, A's payoff as a fraction of the maximal joint value W^* is:

A. $1 - \delta_A(1 - \delta_S)$

B. $(1 - \delta_S)/(1 - \delta_A\delta_S)$

C. $\delta_A(1 - \delta_S)$

D. $(1 - \delta_A)/(1 - \delta_A\delta_S)$

E. $1 - \delta_S(1 - \delta_A)$

b) (3 points) In regime I with an infinite horizon, let $\delta_i = e^{-r_i \Delta}$, where $r_i > 0$ is i's discount rate and Δ the time between offers. As $\Delta \to 0$, A's equilibrium payoff converges to the generalized Nash solution (threat points 0) in which A's weight β_A equals:

A. $r_A/(r_A + r_S)$

B. $1/2$

C. $r_S/(r_A + r_S)$

D. r_S/r_A

E. $r_A r_S/(r_A + r_S)^2$

c) (2 points, T or F) In regime I with an infinite horizon, stronger complementarity (higher b) raises A's equilibrium payoff while leaving A's share of W^* unchanged.

d) (2 points, T or F) In regime II, for every $c > b > 0$ and every $\sigma \in (0, 1)$, raising A's revenue share σ lowers S's equilibrium level y^N.

Question 6. (15 points) *Integrative: complementarities (Aula 1) × value maximization and the Coase theorem (Aula 2) × alternating-offers bargaining (MWG 9.AA).* Use the common setting above.

a) (3 points) Write the joint value $W(x, y)$, give the condition under which it has a unique maximizer, and find the efficient pair and W^*.

Resolution:

b) (3 points) Regime I, infinite horizon. Characterize the stationary subgame-perfect equilibrium: show that every equilibrium offer specifies (x^*, y^*) whatever δ_A, δ_S; derive both payoffs and the transfer τ; say when agreement occurs.

Resolution:

c) (3 points) Regime II. Derive the Nash equilibrium (x^N, y^N) as a function of σ and prove it is unique. Prove that no $\sigma \in [0, 1]$ makes it efficient. For $\sigma = \frac{1}{2}$, draw in the (x, y) plane the two equilibrium best replies, the two "efficient" reaction lines $x = (t + by)/c$ and $y = (t + bx)/c$, and both intersection points.

Resolution:

d) (3 points) At $\sigma = \frac{1}{2}$, find in closed form the efficiency loss $L = W^* - W(x^N, y^N)$ and the relative loss L/W^*. Prove that the relative loss is increasing in b and find its limits as $b \to 0$ and $b \to c$.
Resolution:

e) (3 points) Parameter analysis. Prove the signs of the effects of t, b, c, δ_A, δ_S on x^*, W^*, A's share s_A and payoff in regime I, and the relative loss L/W^* in regime II at $\sigma = \frac{1}{2}$. Tabulate them, study $\delta_A = \delta_S = \delta \to 1$, $\delta_S \to 0$ and $b \to c$, and explain why the choice between regimes matters more when the investments are strong complements.
Resolution: